

Calculations

System of Units

- Length: Kilometers (km)
- Mass: Solar Masses ($M_{\odot}$)
- Time: Seconds (s)
- Energy Density/Pressure: Megaelectronvolts per cubic femtometer (MeV · fm³)

Relativistic Units

Many of the calculations utilize a relativistic system of units where the speed of light (c) is a dimensionless value equal to 1. This leads to the ability to express length as a unit of time. Further reading.

$$c = L \cdot T^{-1}$$

$$1 = L \cdot T^{-1}$$

$$T = L$$

Radius Scaling

The relativistic, dimensionless scaling for radius is (see TOV.md)

$$r = a \cdot r'$$

$$[r] = [a] \cdot [r']$$

Since r' is dimensionless, $[r'] = 1$. Thus, $[r] = [a]$

$$a = (G \cdot \varepsilon_0)^{-1/2}$$

$$a = G^{-1/2} \cdot \varepsilon_0^{-1/2}$$

$$[a] = [G]^{-1/2} \cdot [\varepsilon_0]^{-1/2}$$

$$[G] = L^3 \cdot M^{-1} \cdot T^{-2} \text{ and } [\varepsilon_0] = L^{-1} \cdot M \cdot T^{-2}$$

$$[a] = (L^3 \cdot M^{-1} \cdot T^{-2})^{-1/2} \cdot (L^{-1} \cdot M \cdot T^{-2})^{-1/2}$$

$$[a] = L^{-3/2} \cdot M^{1/2} \cdot T \cdot L^{1/2} \cdot M^{-1/2} \cdot T$$

$$[a] = L^{-1} \cdot T^2$$

Since a has the same dimension as radius, it should be a unit of length ($[a] = L$).
However

$$[a] = L \cdot (L^{-2} \cdot T^2)$$

For a to represent a physical quantity, it must be multiplied by a quantity with a dimension of $L^2 \cdot T^{-2}$. In this case, that quantity is c , specifically c^2 (since that's what everything is scaled by).

$$\begin{aligned} r_{rel} &= a_{rel} \cdot r' \\ r_{phys} &= r_{rel} \cdot c^2 \end{aligned}$$

As long as G and ε_0 are using the same system of units, r_{phys} will be in the base unit of length. For the sake of simplicity, the project just uses SI units. Therefore, r_{phys} is in units of meters. To convert to kilometers, simply divide by 1000: $r_{km} = r_{phys} \div 1000$.

Mass Rescaling

The process for rescaling mass is almost identical to the process for rescaling radius.

$$\begin{aligned} m_r &= b \cdot m'_r \\ [m_r] &= [b] \end{aligned}$$

By definition

$$\begin{aligned} b &= (G^3 \cdot \varepsilon_0)^{-1/2} \\ b &= G^{-3/2} \cdot \varepsilon_0^{-1/2} \end{aligned}$$

Therefore

$$\begin{aligned} [b] &= (L^3 \cdot M^{-1} \cdot T^{-2})^{-3/2} \cdot (L^{-1} \cdot M \cdot T^{-2})^{-1/2} \\ [b] &= L^{-9/2} \cdot M^{3/2} \cdot T^3 \cdot L^{1/2} \cdot M^{-1/2} \cdot T \\ [b] &= L^{-4} \cdot M \cdot T^4 \end{aligned}$$

b should have a dimension of mass, so it must be multiplied by c^4 .

$$\begin{aligned} m_{rel} &= b \cdot m'_r \\ m_{phys} &= m_{rel} \cdot c^4 \end{aligned}$$

As long as G and ε_0 are using the same system of units, m_{phys} will be in the base unit of mass. For the sake of simplicity, the project just uses SI units. Therefore, m_{phys} is in units of kilograms. To convert to solar masses, simply divide by the mass of the sun. $m_{M\odot} = m_{kg} \div (1.988416 \times 10^{30})$.